RTCA, Inc.

1828 L Street, NW Suite 805

Washington, DC 20036

RTCA/DO-160G

Environmental Conditions and Test Procedures for Airborne Equipment

Section 12

Sand and Dust

Important Notice

Information contained in these sections is pertinent to all test procedures described in the other sections of this document. Further, Appendix A is applicable for identifying environmental tests performed.

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12.0 Sand and Dust

12.1 Purpose of the Test

This test determines the resistance of the equipment to the effects of blowing sand and dust where carried by air movement at moderate speeds. The main adverse effects to be anticipated are:

a. Penetration into cracks, crevices, bearings and joints, causing fouling and/or clogging of moving parts, relays, filters, etc.

b. Formation of electrically conductive bridges.

c. Action as nucleus for the collection of water vapor, including secondary effects of possible corrosion.

d. Pollution of fluids.

NOTE: Consideration must be given in determining where in the sequence of environmental tests to apply this test procedure, as dust residue from this test procedure, combined with other environmental synergistic effects may corrode or cause mold growth on the test item and adversely influence the outcome of succeeding test procedures. Sand abrasion may also influence the results of the salt spray, fungus or humidity test procedures.

12.2 Categories of Equipment

Category D

Equipment tested as recommended in the following paragraphs for Dust test is identified as Category D. Such equipment can be installed in locations where the equipment is subjected to blowing dust in the course of normal aircraft operations.

Category S

Equipment tested as recommended in the following paragraphs for both Dust test and Sand test is identified as Category S. Such equipment can be installed in locations where the equipment is subjected to blowing sand and dust in the course of normal aircraft operations. Such location includes cockpit or any other location not intentionally protected against sand and dust exposure.

12.3 Agent

12.3.1 Dust

Dust used in a suitable test chamber vented to the atmosphere shall be raised and maintained at a concentration of 3.5 to 8.8 g/m3 and shall be 97% to 99% silicon dioxide.

Dust to be used can be Red china clay or Silica Flour and must meet size distributions of 100% by weight less than 150 μm, with a median diameter (50 ±2 % by weight) of 20 ±5 μm. This dust is readily available as a 140 mesh Silica Flour (about 2% retained on a 140 mesh (108 microns) sieve) and should provide comparable results to prior test requirements. National documentation may contain other more specific distributions.

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12.3.2 Sand

(1) If sand concentration data is available for the worst case field environment the equipment is expected to properly function in, then test to that sand concentration level.

(2) For materiel likely to be used close to helicopters operating over unpaved surfaces: 2.2 +/- 0.5 g/m3 (0.06 +/- 0.015 g/per cubic foot).

(3) For material never used or exposed in the vicinity of operating aircraft, but which may be used or stored unprotected near operating surface vehicles: 1.1 +/- 0.3 g/m3 (0.033 +/- 0.0075 g/per cubic foot).

(4) For material that will be subjected only to natural conditions: 0.18 g/m3, -0.0/+0.2 g/m3 (0.005 g/per cubic foot). (This large tolerance is due to the difficulties of measuring concentrations at low levels.).

Unless otherwise specified, for the sand test use silica sand (at least 95 percent by weight SiO2). Use sand of sub-angular structure, a mean Krumbein number range of 0.5 to 0.7 for both roundness and sphericity and a hardness factor of 7 mhos. Due to the loss of subangular structure and contamination, re-use of test sand is normally not possible. The recommended particle size distribution for the large particle sand test is from 150 μm to 850 μm, with a mean of 90 ±5 percent by weight smaller than 600 μm and larger than or equal to 150 μm, and at least 5 percent by weight 600 μm and larger.

12.4 Dust Test Procedure

The equipment shall be submitted to the dust test along each direction of each major orthogonal axis in succession. The air velocity shall be maintained between 0.5 and 2.4 m/second.

Note 1: *Unless otherwise required in the relevant specification, the equipment is not required to operate during the exposure period.*

Note 2*: Health and safety regulations regarding the use of dust should be observed.*

12.4.1 First Cycle

With the internal temperature of the test chamber maintained at +25 ±2 degrees Celisus (°C) and the relative humidity at not more than 30 percent, submit the equipment to a minimum exposure period of one hour along each direction of each major orthogonal axis in succession.

12.4.2 Second Cycle

With the internal temperature of the test chamber raised and stabilized at +55 ±2 °C and the relative humidity at not more than 30 percent, submit the equipment to a minimum exposure period of one hour along each direction of each major orthogonal axis in succession.

At the end of this exposure period, the equipment shall be removed from the chamber and cooled to room temperature. Externally accumulated dust only on surfaces of the equipment required to verify proper operation (e.g. displays, connectors, keyboards, test ports etc.) shall be removed by brushing, wiping or shaking with care being taken to avoid introducing additional dust into the equipment. Under no circumstances shall dust

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be removed by either air blast or vacuum cleaning. After removing the excess dust, manipulate the mechanical moving features of the equipment ten times and check for clogging or binding of movement. DETERMINE COMPLIANCE WITH APPLICABLE EQUIPMENT PERFORMANCE STANDARDS.

12.5 Sand Test Procedure

The test item shall be submitted to the sand test at wind velocities ranging from 18-29 m/s, with the exposed surface positioned 3-m from the sand injection point - to allow sand particles to reach terminal wind velocities before impacting the test item. If the test item will not directly see wind velocities of this magnitude during normal service life (cockpit screen displays) then the test item shall be submitted to air velocities of 0.5 and 2.4 m/second. A vertical chamber may be utilized to preserve the uniformity of sand distribution.

Note 1: Unless otherwise required in the relevant specification, the equipment is not required to operate during the exposure period.

12.5.1 First Cycle

With the internal temperature of the test chamber maintained at +25 ±2 °C and the relative humidity at not more than 30 percent, submit the equipment to a minimum exposure period of one hour along each direction of each major orthogonal axis in succession.

12.5.2 Second Cycle

With the internal temperature of the test chamber raised and stabilized at +55 ±2 °C and the relative humidity at not more than 30 percent, submit the equipment to a minimum exposure period of one hour along each direction of each major orthogonal axis in succession.

At the end of this exposure period, the equipment shall be removed from the chamber and cooled to room temperature. Externally accumulated sand only on surfaces of the equipment required to verify proper operation (e.g. displays, connectors, keyboards, test ports etc.) shall be removed by brushing, wiping, or shaking with care being taken to avoid introducing additional sand into the equipment. Under no circumstances shall sand be removed by either air blast or vacuum cleaning. After removing the excess sand, manipulate the mechanical moving features of the equipment ten times and check for clogging or binding of movement. DETERMINE COMPLIANCE WITH APPLICABLE EQUIPMENT PERFORMANCE STANDARDS.

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